

CO2 Assessment Tool 2

3. IIR Filter Coefficient Quantization

Simulate a second-order Butterworth IIR low-pass filter with 10-bit coefficient quantization and analyze the effects of finite word length on filter performance. (25) (BL4) (WK4)

Given Data:

- Filter Type: Butterworth
- Order = 2
- Cutoff Frequency = 0.3π
- Coefficient Word Length = 10 bits
- Sampling Frequency = 10 kHz

Tasks:

1. Design the Butterworth filter.
2. Determine the floating-point coefficients.
3. Quantize coefficients to 10-bit precision.
4. Plot pole-zero diagrams before and after quantization.
5. Analyze stability changes.
6. Compare frequency responses.

MATLAB CODE

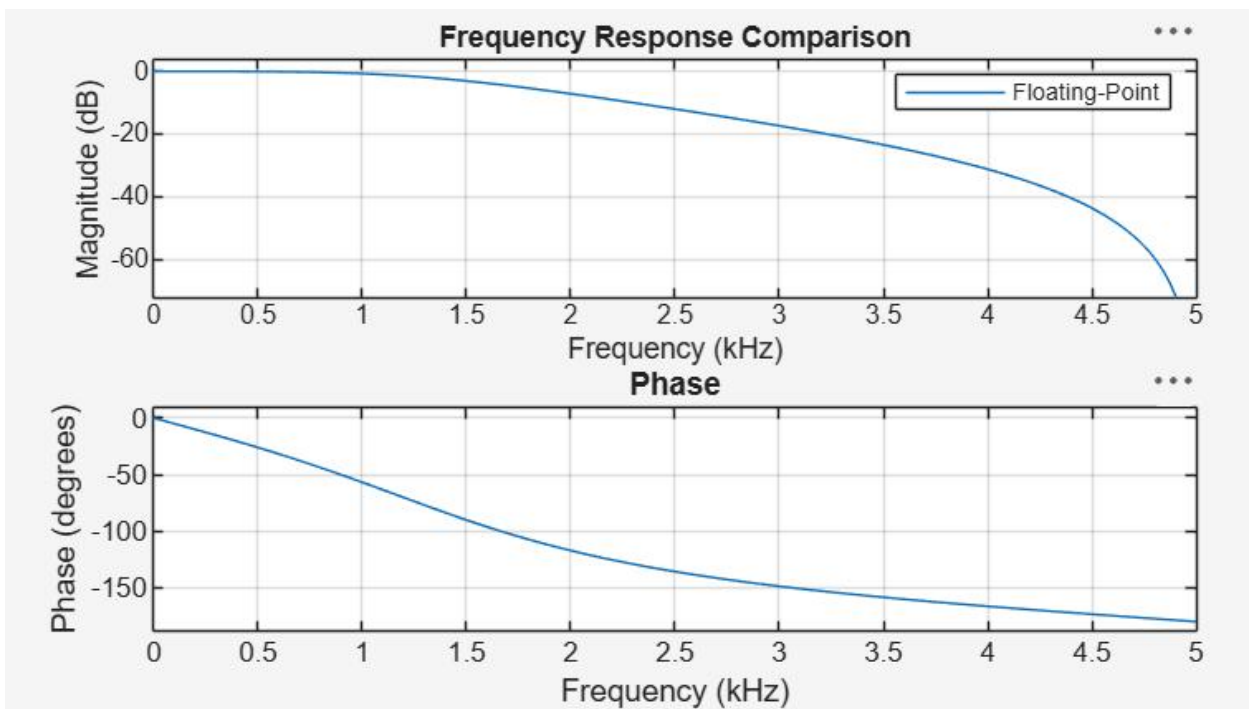
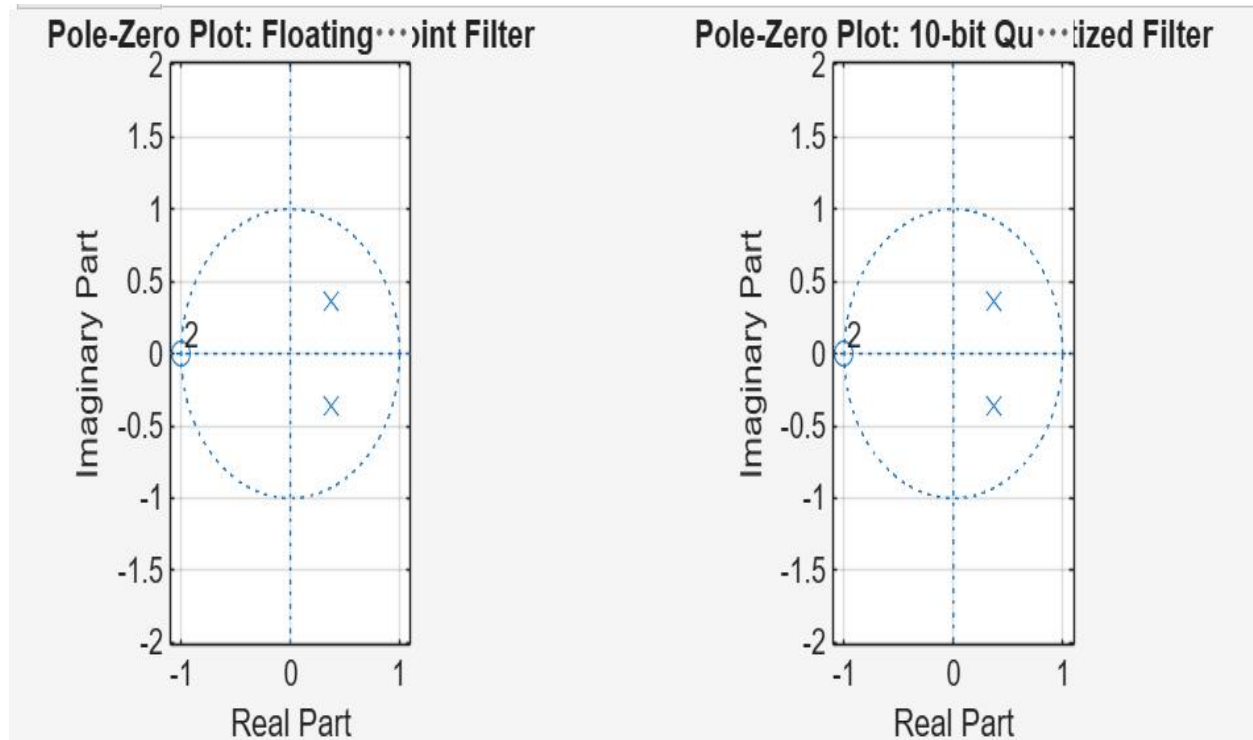
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clc;
clear;
close all;
%% Given Parameters
Fs = 10000;           % Sampling frequency
Wn = 0.3;             % Normalized cutoff frequency
Nbits = 10;           % Coefficient word length
%% 1. Design 2nd Order Butterworth Low-Pass Filter
[b,a] = butter(2,Wn,'low');
disp('Floating-point numerator coefficients:');
disp(b);
disp('Floating-point denominator coefficients:');
disp(a);
%% 2. Quantization to 10-bit fixed-point
F = 9;                % Fractional bits for Q1.9
scale = 2^F;
bq = round(b*scale)/scale;
aq = round(a*scale)/scale;
disp('Quantized numerator coefficients:');
disp(bq);
disp('Quantized denominator coefficients:');
disp(aq);
%% 3. Pole-Zero Calculation
z1 = roots(b);
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p1 = roots(a);
z2 = roots(bq);
p2 = roots(aq);
disp('Floating-point poles:');
disp(p1);
disp('Quantized poles:');
disp(p2);
%% 4. Pole-Zero Diagrams
figure;
subplot(1,2,1);
zplane(b,a);
title('Pole-Zero Plot: Floating-Point Filter');
grid on;
subplot(1,2,2);
zplane(bq,aq);
title('Pole-Zero Plot: 10-bit Quantized Filter');
grid on;
%% 5. Stability Analysis
pole_radius_original = abs(p1);
pole_radius_quantized = abs(p2);
disp('Original pole magnitudes:');
disp(pole_radius_original);
disp('Quantized pole magnitudes:');
disp(pole_radius_quantized);
if all(pole_radius_original < 1)
    disp('Original filter is STABLE');
else
    disp('Original filter is UNSTABLE');
end
if all(pole_radius_quantized < 1)
    disp('Quantized filter is STABLE');
else
    disp('Quantized filter is UNSTABLE');
end
%% 6. Frequency Response Comparison
figure;
freqz(b,a,1024,Fs);
hold on;
freqz(bq,aq,1024,Fs);
legend('Floating-Point','10-bit Quantized');
title('Frequency Response Comparison');
grid on;

```

OUTPUT



Floating-point numerator coefficients:
0.1311 0.2622 0.1311

Floating-point denominator coefficients:
1.0000 -0.7478 0.2722

Quantized numerator coefficients:
0.1309 0.2617 0.1309

Quantized denominator coefficients:
1.0000 -0.7480 0.2715

Floating-point poles:
 $0.3739 + 0.3639i$
 $0.3739 - 0.3639i$

Quantized poles:
 $0.3740 + 0.3628i$
 $0.3740 - 0.3628i$

Original pole magnitudes:
0.5217
0.5217

Quantized pole magnitudes:
0.5210
0.5210

Original filter is STABLE
Quantized filter is STABLE